Exam Ace Champs
Where Champions Are Made

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Chapterwise

Mind-Maps

MATHEMATICS

class 12

Covers
all chapters of NCERT

Faster Recall

Quick Revision

Relations and Functions

Composition of functions and Invertible functions

Invertible functions

The composition of functions $f : A \to B$ and $g : B \to C$ is denoted by gof, and is defined as $gof : A \to C$ given by $gof(x) = g(f(x))\ \forall x \in A$. *e.g.* let $A = N$ and $f, g : N \to N$ such that $f(x) = x^2$ and $g(x) = x^3\ \forall\ x \in N$. Then $gof(2) = g(f(2)) = g(2^2) = 4^3 = 64$.

Composition of functions

A function $f : x \to y$ is invertible, if $\exists$ a function $g : y \to x$ such that $gof = I_x$ and $fog = I_y$. Then, g is the inverse of f. If f is invertible, then it is both one-one and onto and vice-versa. For *eg*. If $f(x) = x$ and $f : N \to N$, then f is invertible.

Theorem 1 : If $f : x \to y$, $g : y \to z$ and $h : z \to s$ are functions, then $ho(got) = (hog)of$.

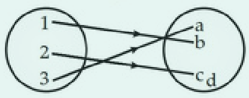
Theorem 2 : Let $f : x \to y$ and $g : y \to z$ be two invertible functions, then gof is invertible and $(gof)^{-1} = f^{-1} o\, g^{-1}$.

Types of functions

One - one (injective)

$f : x \to y$ is one-one if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2\ \forall\ x_1, x_2 \in x$. Other wise, f is many- one, f_1 is one-one.

f_1



Onto (Surjective)

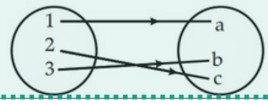
$f : x \to y$ is onto it for energy $y \in Y, \exists\ x \in X$ S.t. $f(x) = y$, f_2 is onto.

f_2

Bijective

$f : x \to y$ is both one-one and onto, then f is bijective. f_3 is bijective.

f_3



Binary Operations Definition and its types

A binary operation '*' on a set A is a function $* : A \times A \to A$ denoted by $a * b$ *i.e.* $\forall\ a, b \in A$, $a * b \in A$. Commutative if $a * b = b * a\ \forall\ a, b \in A$. Associative if $(a * b) * c = a * (b * c)\ \forall a, b \in A$. $e \in A$ is identity if $a * e = a = e * a\ \forall a \in A$. and $b \in A$ is the inverse of $a \in A$, if. $a * b = e = b * a$. Addition is a binary operation on the set of integers.

Types of Relations

Empty Relations Universal relation

A relation $R : A \to A$ is empty if $a \not R b\ \forall\ a, b \in A$. $R = \phi \subset A \times A$.
For *eg* : $R = \{(a,b) : a = b^2\}$, $A = \{1,5,10\}$

A relation $R : A \to A$ is universal if $a\ R\ b\ \forall, a,b \in A, R = A \times A$.
if $R = \phi$, then R is universal.

Trivial Relations

Reflexive relation

A relation $R : A \to A$ is reflexive if $aRa\ \forall\ a \in A$

Symmetric Relation

A relation $R : A \to A$ is symmetric if $aRb \Rightarrow bRa\ \forall\ a, b \in A$

Transitive relation

A relation $R : A \times A$ is transitive if $aRb,\ bRc \Rightarrow aRc\ \forall\ a,b,c \in A$.

Equivalence relation
(reflexive, symmetric, transitive *e.g.*, Let T = the set of all triangles in a plane and $R : T \to T$ defined by $R = \{(T_1, T_2)\} : T_1$ is congruent to $T_2\}$. Then, R is equivalence.

Inverse Trigonometric Functions

Domain and range of inverse trigonometric functions

(i) $y = \sin^{-1}x$. Domain $= [-1,1]$, Range $= \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(ii) $y = \cos^{-1}x$. Domain $= [-1,1]$ Range $= [0, \pi]$

(iii) $y = \operatorname{cosec}^{-1}x$. Domain $= R-(-1,1)$, Range $= \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]-\{0\}$

(iv) $y = \sec^{-1}x$. Domain $= R-(-1,1)$, Range $= [0, \pi] - \left\{\frac{\pi}{2}\right\}$

(v) $y = \tan^{-1}x$. Domain $= R$, Range $= \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(vi) $y = \cot^{-1}x$. Domain $= R$, Range $= (0, \pi)$.

Trigonometric functions

(I) $\sin : R \to [-1,1]$

(ii) $\cos : R \to [-1,1]$

(iii) $\tan : R-\left\{x : x = (2n+1)\frac{\pi}{2} in \in Z\right\} \to R$

(iv) $\cot : R - \{x : x = h\pi, n \in Z\} \to R$

(v) $\sec : R - \left\{x : x = (2n+1)\frac{\pi}{2}, n \in Z\right\} \to R-(-1,1)$

(vi) $\operatorname{cosec} : R - \{x : x = h\pi, n \in Z\} \to R-(-1,1)$

Principal Value branch and principal value

The range of an inverse trigonometric function is the principal value branch and those values which lies in the principal value branch is called the principal value of that inverse trigonometric functions.

Some important relations

$\sin^{-1}x \neq (\sin x)^{-1}.(\sin x)^{-1}$
$= \frac{1}{\sin x}$ and same for other trigonometric functions.

(i) $y = \sin^{-1}x \Rightarrow x = \sin y$

(ii) $x = \sin y \Rightarrow y = \sin^{-1}x$

(iii) $\sin(\sin^{-1}x) = x$

(iv) $\sin^{-1}(\sin x) = x$

(v) $\sin^{-1}\frac{1}{x} = \operatorname{cosec}^{-1}x$

(vi) $\cos^{-1}(-x) = \pi - \cos^{-1}x$

(vii) $\cos^{-1}\frac{1}{x} = \sec^{-1}x$

(viii) $\cot^{-1}(-x) = \pi - \cot^{-1}x$

(ix) $\tan^{-1}\frac{1}{x} = \cot^{-1}x$

(x) $\sec^{-1}(-x) = \pi - \sec^{-1}x$

(xi) $\sin^{-1}(-x) = -\sin^{-1}x$

(xii) $\tan^{-1}(-x) = -\tan^{-1}x$

(xiii) $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$

(xiv) $\operatorname{cosec}^{-1}x + \sec^{-1}x = \frac{\pi}{2}$

(xv) $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\frac{x+y}{1-xy}$

(xvi) $2\tan^{-1}x = \tan^{-1}\frac{2x}{1-x^2}$

(xvii) $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\frac{x-y}{1+xy}$

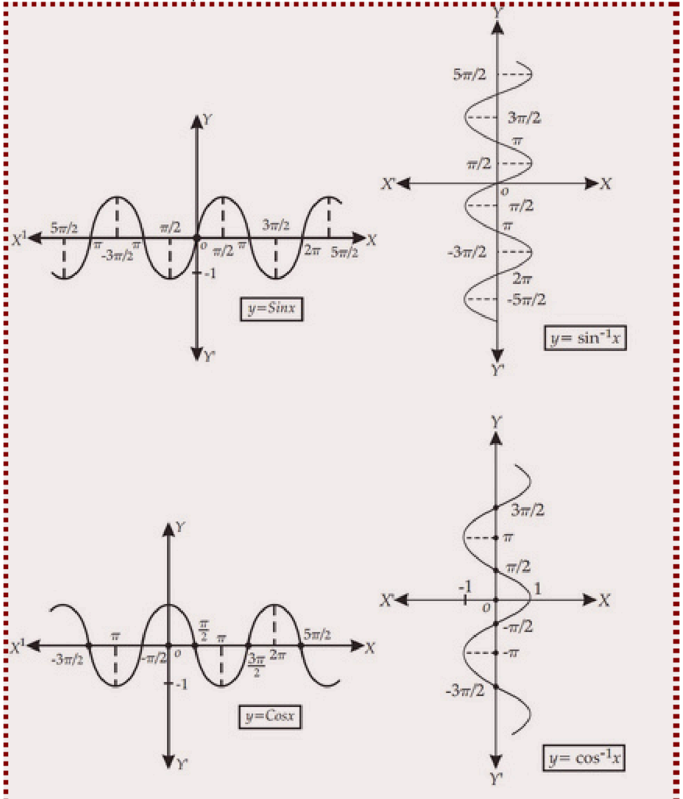
(xviii) $2\tan^{-1}x = \sin^{-1}\frac{2x}{1+x^2} = \cos^{-1}\frac{1+x^2}{1-x^2}$

For *eg* : to find the principal value of $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$, let $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = y$,

$\Rightarrow \sin y = \frac{1}{\sqrt{2}}$ The range or the principal value branch of $\sin^{-1}$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

and $\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$ So, the principal value of $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ is $\frac{\pi}{4}$

Graphs of trigonometric functions and inverse trigonometric functions



If $A=[a_{ij}]_{m\times n}$, then its transpose A' $(A^T)=[a_{ji}]_{n\times m}$ i.e. if $A=\begin{pmatrix}2 & 1\end{pmatrix}$ then $A^T=\begin{pmatrix}1\\2\end{pmatrix}$.

Also, $(A')'=A$, $(kA)'=kA'$, $(A+B)'=A'+B'$, $(AB)'=B'A'$.

- A is symmetric matrix if $A=A'$ i.e. $A'=-A$.
- A is skew – symmetric if $A=-A'$ i.e. $A'=-A$.
- A is any matrix, then–

$$A=\frac{1}{2}\{\underbrace{(A+A')}_{\text{S.M}}+\underbrace{(A-A')}_{\text{Skew, S.M.}}\}=\text{sum of a symmetric and a skew-symmetric matrix.}$$

For eg if $A=\begin{pmatrix}2 & 8\\6 & 4\end{pmatrix}$, then $A=\frac{1}{2}\left\{\begin{pmatrix}2 & 7\\7 & 4\end{pmatrix}+\begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix}\right\}$.

A matrix of order m×n is an ordered rectangular array of numbers or functions having 'm' rows and 'n' columns. The matrix $A=[a_{ij}]_{m\times n}$ is given by

$$A=\begin{bmatrix}a_{11} & a_{12} & \dots a_{1n}\\a_{21} & a_{22} & \dots a_{2n}\\a_{m1} & a_{m2} & \dots a_{mn}\end{bmatrix}_{m\times n}$$

- Column matrix : It is of the form $[a_{ij}]_{m\times 1}$
- Row matrix : It is of the form $[a_{ij}]_{1\times n}$
- Square matrix : Here, $m=n$ (no. of rows = no. of columns)
- Diagonal matrix : All non-diagonal entries are zero i.e $a_{ij}=0\ \forall i\neq j$
- Scalar matrix : $a_{ij}=0, i\neq j$ and $a_{ij}=k\,(\text{Scalar}), i=j$
- Identity matrix : $a_{ij}=0, i\neq j$ and $a_{ij}=1, i=j$
- Zero matrix : All entries are zero.

Types of Matrix

Definition and its types

$A=[a_{ij}]=[b_{ij}]=B$ if, A and B are of same order and $a_{ij}=b_{ij}\ \forall\ i$ and j.

Elementary operations on a matrix

Matrices

Equality of Two Matrix

$R_i\leftrightarrow R_j$ or $C_i\leftrightarrow C_j$

$R_i\to kR_i$ or $C_i\to kC_i$

$R_i\to R_i+kR_j$ or $C_i\to C_i+kC_j$

Operations on matrices

If A, B are two matrices of same order, then $A+B=[a_{ij}+b_{ij}]$ The addition of A and B follows: $A+B=B+A$, $(A+B)+C=A+(B+C)$, $-A=(-1)A$, $k(A+B)=kA+kB$, k is scalar and $(k+l)A=kA+lA$, k and l are constants.

If A, B are square matrices such that $AB=BA=I$ then $B=A^{-1}$ i.e., A is the inverse of B and vice-versa.

Inverse of a square matrix, if it exists, is unique

For eg: If $A=\begin{pmatrix}1 & 2\\3 & 4\\5 & 6\end{pmatrix}$, then after $R_1\leftrightarrow R_2$, A becomes $\begin{pmatrix}3 & 4\\1 & 2\\5 & 6\end{pmatrix}$

If A and B are invertible matrices of the same order, then $(AB)^{-1}=B^{-1}A^{-1}$. By elementary transformations, we can convert $A=IA$ to $I=A^{-1}A$. This is one process of finding the inverse of a given square matrix A.

Addition

- If $A=\begin{pmatrix}2 & 3\\2 & 4\end{pmatrix}$, $B=\begin{pmatrix}-3 & 2\\-4 & 5\end{pmatrix}$ then $A+B=\begin{pmatrix}-1 & 5\\-2 & 9\end{pmatrix}$
- If $A=(2\ 3)_{1\times 2}$, $B=\begin{pmatrix}4\\5\end{pmatrix}_{2\times 1}$, then $AB=(2\times 4+3\times 5)=(2\ 3)_{1\times 1}$

Multiplication

If $A=[a_{ij}]_{m\times n}$ and $B=[b_{jk}]_{n\times p}$, then $AB=C=[C_{ik}]_{m\times p}$, $[C_{jk}]=\sum_{j=1}^{n}a_{ij}b_{jk}$. Also, $A(BC)=(AB)C$, $A(B+C)=AB+AC$ and $(A+B)C=AC+BC$, but $AB\neq BA$ (always).

Class : 12th Maths
Chapter- 4 : Determinants

Determinants

Minors and cofactors of a matrix

Minor of an element a_{ij} in a determinant of matrix A is the determinant obtained by deleting i^{th} row and j^{th} column and is denoted by M_{ij}. If M_{ij} is the minor of a_{ij} and cofactor of a_{ij} is A_{ij} given by $A_{ij} = (-1)^{i+j} M_{ij}$.

- If $A_{3\times3}$ is a matrix, then $|A| = a_{11}.A_{11} + a_{12}.A_{12} + a_{13}.A_{13}$.
- If elements of one row (or column) are multiplied with cofactors of elements of any other row (or column), then their sum is zero. For *e.g.*, $a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{33} = 0$.

e.g., if $A = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}$, then $M_{11} = 4$ and $A_{11} = (-1)^{1+1}4 = 4$.

Determinant of square matrix 'A' |A| is given by

(i) if $A = [a_{11}]_{1\times1}$, then $|A| = a_{11}$

(ii) if $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}_{2\times2}$ then $|A| = a_{11}\,a_{22} - a_{12}\,a_{21}$

(iii) if $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3\times3}$, then $|A| = a_{11}(a_{22}.a_{33} - a_{23}.a_{32}) - a_{12}(a_{21}.a_{33} - a_{23}.a_{31}) + a_{13}(a_{21}\,a_{32} - a_{22}\,a_{31})$

For *eg.* if $A = \begin{bmatrix} 2 & 3 \\ 2 & 4 \end{bmatrix}$, then $|A| = 2 \times 4 - 3 \times 2 = 2$

Properties of |A|

(i) $|A|$ remains unchanged, if the rows and columns of A are interchanged ie., $|A| = |A'|$

(ii) if any two rows (or columns) of A are interchanged, then the sign of $|A|$ changes.

(iii) if any two rows (or columns) of A are identical, then $|A| = 0$

(iv) if each element of a row (or a column) of A is multiplied by B (const.), then $|A|$ gets multiplied by B.

(v) if $A = [a_{ij}]_{3\times3}$ then $|k.A| = k^3|A|$.

(vi) if elements of a row or a column in a determinant $|A|$ can be expressed as sum of two or more elements, then $|A|$ can be expressed as $|B| + |C|$.

(vii) if $R_i \to R_i + kR_j$ or $C_i = C_i + kC_j$ in $|A|$, then the value of $|A|$ remains same

Adjoint and inverse of a matrix

if $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, then adj. $A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}$

, where A_{ij} is the cofactor of a_{ij}.

- $A(\text{adj}.A) = (\text{adj}.A).A = |A| = I$, A – square matrix of order 'n'
- if $|A| = 0$, then A is singular. Otherwise, A is non-singular.
- if $AB = BA = I$, where B is a square matrix, then B is called the inverse of A, $A^{-1} = B$ or $B^{-1} = A$, $(A^{-1})^{-1} = A$.

Inverse of a square matrix exists if A is non-singular *i.e.* $|A| \neq 0$, and is given by

$$A^{-1} = \frac{1}{|A|}(\text{adj}.A)$$

Area of triangle

if $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) $\Delta = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

For eg: if (1, 2) , (3, 4) and (–2, 5) are the vertices, then area of the triangle is

$$\Delta = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 4 & 1 \\ -2 & 5 & 1 \end{vmatrix} = 1(4-5) - 2(3+2) + 1(15+8) = 12 \text{ sq.units.}$$

we take positive value of the determinant.

Applications of Determinants

- if $a_1x + b_1y + c_1z = d_1$, $a_2x + b_2y + c_2z = d_2$, $a_3x + b_3y + c_3z = d_3$ then we can write $AX = B$,

where $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$

- Unique solution of $AX = B$ is $X = A^{1}B$, $|A| \neq 0$.
- $AX = B$ is consistent or inconsistent according as the solution exists or not.
- For a square matrix A in $AX = B$, if
  (i) $|A| \neq 0$ then there exists unique solution.
  (ii) $|A| = 0$ and (adj. A) $B \neq 0$,then no solution.
  (iii) if $|A| = 0$ and $(\text{adj}.A).B = 0$ then system may or may not be consistent.

Class : 12th Maths
Chapter- 5 : Continuity and Differentiability

Continuity and Differentiability

Continuous Function

Suppose f is a real function on a subset of the real numbers and let 'c' be a point in the domain of f.
Then f is continuous at c if $\lim_{x\to c} f(x) = f(c)$
A real function f is said to be continuous if it is continuous at every point in the domain of f. For eg: The function $f(x)=\frac{1}{x}, x\neq 0$ is continuous
Let 'C' be any non-zero real number, then $\lim_{x\to c} f(x) \lim_{x\to c}\frac{1}{x} = \frac{1}{c}$. For $c=0, f(c)=\frac{1}{c}$ So $\lim_{x\to c} f(x) = f(c)$ and hence f is continuous at every point in the domain of f.

Derivatives of functions in parametric form

Let $x = f(t)$, $y = g(t)$ be two functions of parameter 't'

Then, $\frac{dy}{dt} = \frac{dy}{dx}\cdot\frac{dx}{dt}$ or $\frac{dy}{dx} = \dfrac{\frac{dy}{dt}}{\frac{dx}{dt}}\left(\frac{dx}{dt}\neq 0\right)$

Jhus, $\frac{dy}{dx} = \frac{g'(t)}{f'(t)}$ [provided $f'(t)\neq 0$]

For eg : if $x = a\cos\theta$, $y = a\sin\theta$ then $\frac{dx}{d\theta} = -a\sin\theta$ and $\frac{dy}{d\theta} = a\cos\theta$, and so $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = -\frac{a\cos\theta}{a\sin\theta} = -\cot\theta$.

Differentiability

Suppose f is a real function and c is a point in its domain. The derivative of f at c is $f'(c) = \lim_{h\to 0}\frac{f(c+h)-f(c)}{h}$
Every differentiable function is continuous, but the converse is not true.

Second order derivative

Let $y = f(x)$ then $\frac{dy}{dx} = f'(x)$, if $f'(x)$ is differentiable, then $\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}f'(x)$ i.e.,
$\frac{d^2y}{dx^2} = f''(x)$ is the second order derivative of y w.r.t. x.
For eg : if $y = 3x^2+2$, then $y' = 6x$ and $y'' = 6$.

Algebra of continuous functions

Suppose f and g are two real functions continuous at a real number c, then, $f+g$, $f-g$, $f.g$ and $\frac{f}{g}$ are continuous at $x = c\,(g(c)\neq 0)$.

Rolle's theorem

if $f:[a,b]\to R$ is continuous on $[a, b]$ and differentiable on (a, b). Such that $f(a)=f(b)$, then $\exists$ some c in (a, b) s.t. $f'(c)=0$.

Mean Value theorem

if $f:[a,b]\to R$ continuous on $[a, b]$ and differentiable on (a, b).
Then $\exists$ some c in (a, b) such that $f'(c) = \frac{f(b)-f(a)}{b-a}$
e.g. Let $f(x)=x^2$ defined in the interval $[2, 4]$. Since $f(x)=x^2$ is continuous in $[2, 4]$ and differentiable in $(2, 4)$ as $f'(x)=2x$ defined in $(2, 4)$. So,

$$f'(c) = \frac{f(b)-f(a)}{b-a} = \frac{16-4}{4-2} = 6, c\in(2, 4).$$

Chain Rule

if $f = vou$, $t = u(x)$ and if both $\frac{dt}{dx}, \frac{dv}{dt}$ exists, then $\frac{df}{dx} = \frac{dv}{dt}\cdot\frac{dt}{dx}$

Some Standard derivatives

(i) $\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}$ (ii) $\frac{d}{dx}(\cos^{-1}x) = -\frac{1}{\sqrt{1-x^2}}$

(iii) $\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$ (iv) $\frac{d}{dx}(\cot^{-1}x) = -\frac{1}{1+x^2}$

(v) $\frac{d}{dx}(\sec^{-1}x) = \frac{1}{x\sqrt{1-x^2}}$ (vi) $\frac{d}{dx}(\operatorname{cosec}^{-1}x) = -\frac{1}{x\sqrt{1-x^2}}$

(vii) $\frac{d}{dx}(e^x) = e^x$ (viii) $\frac{d}{dx}(\log x) = \frac{1}{x}$

Logarithmic differentiation

$$\text{Let } y = f(x) = [u(x)]^{v(x)}$$

$$\log y = v(x)\log[u(x)]$$

$$\frac{1}{y} = v(x)\frac{1}{u(x)}u'(x) + v'(x)\log[u(x)]$$

$$\frac{dy}{dx} = y\left[\frac{v(x)}{u(x)}\right]u'(x) + v'(x)\log[u(x)]$$

For e.g. : Let $y = a^x$ Then $\log y = x\log a$

$$\frac{1}{y}\cdot\frac{dy}{dx} = \log a$$

$$\frac{dy}{dx} = y\log a = a^x\log a.$$

Applications of Derivatives

Approximations

Let $y=f(x)\Delta x$ be a small increment in 'x' and Δy be the small increment in y corresponding to the increment in 'x', i.e. $\Delta y = f(x+\Delta x) - f(x)$. Then, Δy is given by $dy = f'(x)dx$ or $dy = \left(\frac{dy}{dx}\right)\Delta x$, is a good approximation of Δy when $dx=\Delta x$ is relatively small and denote by $dy \approx \Delta y$. For eg: Let us approximate $\sqrt{36.6}$. To do this, we take $y=\sqrt{x}, x=36$, $\Delta x = 0.6$ then $\Delta y = \sqrt{x+dx} - \sqrt{x}$

$$= \sqrt{36.6} - \sqrt{36}$$

$$= \sqrt{36.6} - 6 \quad \Rightarrow \sqrt{36.6} = 6 + \bar{d}y$$

Now, dy is approximately Δy and is given by $\bar{d}y$

$= \left(\frac{dy}{dx}\right)\Delta x = \frac{1}{2\sqrt{x}}(0.6) = \frac{1}{2\sqrt{36}}(0.6) = 0.05$. So, $\sqrt{36.6} \approx 6 + 0.05 = 6.05$.

Rate of Change Quantities

If a quantity if 'y' varies with another quantity x so that $y = f(x)$, then $\frac{dy}{dx}[f'(x)]$ represents the rate of change of y w.r.t x and $\left.\frac{dy}{dx}\right|_{x=x_0} (f'(x_0))$ represents the rate of change of y w.r.t. x at $x = x_0$

If 'x' and 'y' varies with another variable 't' i.e., if $x = f(t)$ and $y = g(t)$, then by chain rule $\frac{dy}{dx} = \frac{dy}{dt} \Big/ \frac{dx}{dt}$, if $\frac{dx}{dt} \neq 0$.

For eg: if the radius of a circle, r = 5 cm, then the rate of change of the area of a circle per second w.r.t 'r' is –
$\left.\frac{da}{dr}\right|_{r=5} = \frac{d}{dr}(\pi r^2)\Big|_{r=5} = 2\pi r\Big|_{r=5} = 10\pi$

Increasing and decreasing functions

A function f is said to be (i) increasing on (a,b) if $x_1 < x_2$ in $(a,b) \Rightarrow f(x_1) \le f(x_2) \forall x_1 x_2 \in (a,b)$, and (ii) decreasing on (a, b) if $x_1 < x_2$ in $(a,b) \Rightarrow f(x_1) > f(x_2) \forall x_1 x_2 \in (a,b)$

If $f'(x) \ge 0 \forall x \in (a,b)$ then f is increasing in (a,b) and if $f'(x) \le 0 \forall x \in (a,b)$, then f is decreasing in (a,b) For eg: Let $f(x) = x^3 - 3x^2 + 4x, x \in R$, then $f'(x) = 3x^2 - 6x + 4 = 3(x-1)^2 + 1 > 0 \forall x \in R$. So, the function f is strictly increasing on R.

Maximum and Minima

A point C in the domain of 'f' at which either $f'(c)=0$ or is not differentiable is called a critical point of f.

First Derivative test

Let f be continuous at a critical point C in open I. Then **(i)** if $f'(x) > 0$ at every point left of C and $f'(x) < 0$ at every point right of C, then 'C' is a point of local maxima. **(ii)** If $f'(x) < 0$ at every point left of C and $f'(x) > 0$ at every point right of C, then 'C' is a point of local minima.
(iii) If $f'(x)$ does not change sign as 'x' increases through C, then 'C' is called the point of inflection.

Second Derivative test

Let f be a function defined on I and CC–I, f is twice differentiable at C. Then
(i) $x=C$ is a point of local max. If $f'(C)=0$ and $f''(C) < 0$, $f(C)$ is local max. of f.
(ii) $x=C$ is a point of local min if $f'(C)=0$ and $f''(C)>0$. $f(C)$ is local min of f. **(iii)** The test fails if $f'(C)=0$ and $f''(C)=0$

Tangents and Normals

The equation of the tangent at (x_0,y_0), to the curve $y = f(x)$ is given by $(y-y_0) = \left.\frac{dy}{dx}\right]_{(x_0,y_0)}(x - x_0)$ if $\frac{dy}{dx}$ does not exists at (x_0, y_0), then the tangent at (x_0, y_0) is parallel to the y–axis and its equation is $x = x_0$. If tangent to a curve $y = f(x)$ at $x = x_0$ is parallel to x–axis, then $\left.\frac{dy}{dx}\right|_{x=x_0} = 0$.

Equation of the normal to the curve

$y = f(x)$ at (x_0,y_0) is $y - y_0 = -\frac{1}{\left.\frac{dy}{dx}\right|_{(x_0,y_0)}}(x - x_0)$ if $=\frac{dy}{dx}$ at (x_0,y_0) is zero,then equation of the normal is $x = x_0$. If $\frac{dy}{dx}$ at (x_0,y_0) does not exist, then the normal is parallel to x-axis and its equation is $y=y_0$ For eg: Let $y=x^3-x$ be a curve,then the slope of the tangent to $y=x^3-x$ at $x=2$ is $\left.\frac{dy}{dx}\right|_{x=2} = 3x^2-1 = 3.2^2-1 = 11$

Integrals

Integration

It is the inverse of differentiation. Let, $\frac{d}{dx}F(x)=f(x)$. Then $\int f(x)dx=F(x)+c$, 'c' constant of integral. These integrals are called indefinite or general integrals.

Properties of indefinite integrals are

(i) $\int[f(x)+g(x)]dx=\int f(x)dx+\int g(x)dx$, (ii) $\int kf(x)dx=k\int f(x)dx$,

For eg : $\int(3x^2+2x)dx=x^3+x^2+c$ where k is real.

Integration by substitution

The method in which we change the variable to some other variable is called the method of substitution

$\int \tan x dx=\log|\sec x|+c$ $\quad$ $\int \cot x dx=\log|\sin x|+c$

$\int \sec x dx=\log|\sec x+\tan x|+c$ $\quad$ $\int \operatorname{cosec} x dx=\log|\operatorname{cosec} x-\cot x|+c.$

integration of some special functions

(i) $\int\frac{dx}{x^2-a^2}=\frac{1}{2a}\log\left|\frac{x-a}{x+a}\right|+c$ (ii) $\int\frac{dx}{a^2-x^2}=\frac{1}{2a}\log\left|\frac{a+x}{a-x}\right|+c$

(iii) $\int\frac{dx}{x^2+a^2}=\frac{1}{a}\tan^{-1}\frac{x}{a}+c$ (iv) $\int\frac{dx}{\sqrt{x^2-a^2}}=\log\left|x+\sqrt{x^2-a^2}\right|+c$

(v) $\int\frac{dx}{\sqrt{a^2-x^2}}=\sin^{-1}\frac{x}{a}+c$ (vi) $\int\frac{dx}{\sqrt{x^2+a^2}}=\log\left|x+\sqrt{x^2+a^2}\right|+c.$

Some Standard integrals

(i) $\int x^n dx=\frac{x^{n+1}}{n+1}+c, n\neq -1$ like, $\int dx=x+c$

(ii) $\int \cos x\, dx=\sin x+c$ (iii) $\int \sin x\, dx=-\cos x+c$

(iv) $\int \sec^2 x\, dx=\tan x+c$ (v) $\int \operatorname{cosec}^2 x\, dx=-\cot x+c$

(vi) $\int \sec x\tan x\, dx=\sec x+c$ (vii) $\int \operatorname{cosec}x\cot x\, dx=-\operatorname{cosec}x+c$

(viii) $\int\frac{dx}{\sqrt{1-x^2}}=\sin^{-1}x+c$ (ix) $\int\frac{dx}{\sqrt{1-x^2}}=-\cos^{-1}x+c$

(x) $\int\frac{dx}{1+x^2}=\tan^{-1}x+c$ (xi) $\int\frac{dx}{1+x^2}=-\cot^{-1}x+c$

(xii) $\int e^x dx=e^x+c$ (xiii) $\int a^x dx=\frac{a^x}{\log a}+c$

(xiv) $\int\frac{dx}{x\sqrt{x^2-1}}=\sec^{-1}x+c$ (xv) $\int\frac{dx}{x\sqrt{x^2-1}}=-\operatorname{cosec}^{-1}x+c$

(xvi) $\int\frac{1}{x}dx=\log|x|+c$

Some special type of integrals

(i) $\int\sqrt{x^2-a^2}\,dx=\frac{x}{2}\sqrt{x^2-a^2}-\frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right|+c.$

(ii) $\int\sqrt{x^2+a^2}\,dx=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\log\left|x+\sqrt{x^2+a^2}\right|+c$

(iii) $\int\sqrt{a^2-x^2}\,dx=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}+c.$

First fundamental theorem of integral calculus

Let the area function be defined by $A(x)=\int_a^x f(x)dx\,\forall x\geq a$, where f is continuous on $[a,b]$ then $A'(x)=f(x)\forall x\in[a,b]$.

Integration by parts

$$\int f_1(x)f_2(x)dx=f_1(x)f_2(x)dx-\int\left[\frac{d}{dx}f_1(x)\int f_2(x)dx\right]dx$$

Second fundamental theorem of integral Calculus

Let f be a continuous function of x defined on $[a,b]$ and let F be another function such that $\frac{d}{dx}F(x)=f(x)\forall x\in$ domain of f, then $\int_a^b f(x)dx=[F(x)+c]_a^b=F(b)-F(a)$. This is called the definite integral of f over the range $[a,b]$, where a and b are called the limits of integration, a being the lower limit and b be the upper limit.

Integration by partial fractions

A rational function of the form $\frac{P(x)}{Q(x)}(Q(x)\neq 0)=T(x)+\frac{P_1(x)}{Q(x)}$, $P_1(x)$ has degree less than that of $Q(x)$. We can integrate $\frac{P_1(x)}{Q(x)}$ by expressing it in the following forms –

(i) $\frac{px+q}{(x-a)(x-b)}=\frac{A}{x-a}+\frac{B}{x-b}, a\neq b.$

(ii) $\frac{px+q}{(x+a)^2}=\frac{A}{x-a}+\frac{B}{(x-a)^2}$

(iii) $\frac{px^2+qx+r}{(x-a)(x-b)(x-c)}=\frac{A}{x-a}+\frac{B}{x-b}+\frac{C}{x-c}$

(iv) $\frac{px^2+qx+r}{(x-a)^2(x-b)}=\frac{A}{x-a}+\frac{B}{(x-a)^2}+\frac{C}{x-b}$

(v) $\frac{px^2+qx+r}{(x-a)(x^2+bx+c)}=\frac{A}{x-a}+\frac{Bx+C}{x^2+bx+c}$

Example

$$\int_{-\pi/4}^{\pi/4}\sin^2 x\,dx$$
$$=2\int_0^{\pi/4}\sin^2 x\,dx$$
$$=2\int_0^{\pi/4}\left(\frac{1-\cos 2x}{2}\right)dx$$
$$=\int_0^{\pi/4}(1-\cos 2x)dx$$
$$=\left[x-\frac{\sin 2x}{2}\right]_0^{\pi/4}$$
$$=\frac{\pi}{4}-\frac{1}{2}$$

Applications of Integrals

Area between two curves

The area of the region enclosed between two curves $y=f(x), y=g(x)$ and the lines $x=a$, $x=b$ is given by

$$A=\int_a^b [f(x)-g(x)]dx, \text{where } f(x)\geq g(x) \text{ in } [a,b]$$

For eg: To find the area of the region bounded by the two parabolas $y=x^2$ and $y^2=x$

(0,0) and (1,1) are points of intersection of $y=x^2$ and y^2-x and, $y^2=x \Rightarrow y=\sqrt{x}=f(x)$, and $y=x^2=g(x)$, where $f(x)\geq g(x)$ in $[0,1]$.

$$\text{Area, } A=\int_0^1 [f(x)-g(x)]dx$$

$$=\int_0^1 \left[\sqrt{x}-x^2\right]dx$$

$$=\left[\frac{2}{3}x^{3/2}-\frac{x^3}{3}\right]_0^1$$

$$=\frac{2}{3}-\frac{1}{3}=\frac{1}{3} \text{ Sq. units.}$$

if $f(x)\geq g(x)$ in $[a,c]$ and $f(x)\leq g(x)$ in $[c,\ b]$, $a<c<b$, then the area is

$$A=\int_a^c [f(x)-g(x)]dx+\int_c^b [g(x)-f(x)]dx$$

Area under simple curves

The area of the region bounded by the curve $x=f(y)$, $y-axis$ and the lines $y=c$ and $y=d\,(d>c)$ is given by $A=\int_c^d x\,dy$ or $\int_c^d f(y)dy$.

For eg : the area bounded by $x=y^3$, $y-axis$ in the I quadrant and the lines $y=1$ and $y=2$ is

$$A=\int_1^2 x\,dy=\int_1^2 y^3\,dy=\left[\frac{1}{4}y^4\right]_1^2=\frac{1}{4}(2^4-1^4)=\frac{15}{4} \text{ Sq. units}$$

The area of the region bounded by the curve $y=f(x)$, x- $axis$ and the lines $x=a$ and $x=b\,(b>a)$ is given by

$$A=\int_a^b y\,dx \text{ or } \int_a^b f(x)dx.$$

For eg : the area bounded by $y=x^2$, x- $axis$ in I quadrant and the lines $x=2$ and $x=3$ is -

$$A=\int_2^3 y\,dx=\int_2^3 x^2\,dx=\left[\frac{x^3}{3}\right]_2^3=\frac{1}{3}(27-8)=\frac{19}{3} \text{Sq.units.}$$

Class : 12th Maths
Chapter- 9 : Differential Equations

Differential Equations

Definition

An equation involving derivatives of the dependent variable with respect to independent variable (variables) is called a differential equation. If there is only one independent variation, then we call it as an ordinary differential equation. For eg: $2\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^3=0.$

Order of a Differential Equation

It is the order of the highest order derivative occurring in the Differential Equation For eg: the order of $\frac{dy}{dx}=e^x$ is one and order of $\frac{d^2y}{dx^2}+x=0$ is two.

Degree of a Differential Equation

It is defined if the Differential Equations is a polynomial equation in its derivatives, and is defined as the highest power (positive integer only) of the highest order derivative.

For eg: the degree of $\left(\frac{d^2y}{dx^2}\right)^3+\frac{dy}{dx}=0$ is three

Order and degree (if defined) of a D.E. are always positive integers.

Solution of a Differential Equation

A function which satisfies the given Differential Equation is called its solution. The solution which contains as many arbitrary constants as the order of the D.E. is called a general solution and the solution free from arbitrary constants is called particular solution.

For eg: $y=e^x+1$ is a solution of $y''-y'=0$.

Since $y'=e^x$ and $y''=e^x \Rightarrow y''-y'=e^x-e^x=0.$

Formation of Differential Equation

To form a Differential Equation from a given function, we differentiate the function successively as many times as the no. of arbitrary constants in the given function, and then eliminate the arbitrary constants.

For eg: Let the function be $y=ax+b$, then we have to differentiate it two times, since there are 2 arbitrary constants a and b. $\therefore y'=a \Rightarrow y''=0$. Jhus $y''=0$ is the required Differential Equation.

Linear Differential Equation

A Differential Equation of the form $\frac{dy}{dx}+Py=Q$, where P, Q are constants or functions or 'x' only is called a first order linear Differential Equations its solution is $ye^{\int P.dx}=\int Q.e^{\int P.dx}dx+c$. For eg : $\frac{dy}{dx}+3y=2x$ has solution $ye^{\int 3.dx}=\int 2x.e^{\int 3.dx}dx+c \Rightarrow ye^{3x}=2\int xe^{3x}+c.$

Homogeneous Differential Equation

A Differential Equation which can be expressed in the form $\frac{dy}{dx}=f(x,y)$ or $\frac{dx}{dy}=g(x,y)$, where, $f(x,y)$ and $g(x,y)$ are homogeneous functions of degree zero is called a homogenous Differential Equation

For $eg:$ $(x^2+xy)dy=(x^2+y^2)dx$

To solve this, we substitute $\frac{y}{x}=v \Rightarrow y=vx.$

The order of a Differential Equations representing a family of curves is same as the number of arbitrary constants present in the equation corresponding to the family of curves. For eg: Let the family of curves be $y=mx$, $m=$ constant, then, $y'=m$

$y=y'x \Rightarrow y=\frac{dy}{dx}x \Rightarrow x\frac{dy}{dx}-y=0.$

Variable Separation Method

It is used to solve such an equation in which variables can be separated completely. For eg: $y\,dx=x\,dy$ can be solved as $\frac{dx}{x}=\frac{dy}{y}$ Integrating both sides $\log x=\log y+\log c \Rightarrow \frac{x}{y}=c \Rightarrow x=cy$,is the solution.

Vector Algebra

Vectors

A quantity that has both magnitude and direction is called a vector. The distance between the initial and terminal points of a vector is called its magnitude. Magnitude of vector $\overrightarrow{AB}$ is |AB|.

Multiplication of vector by a scalar

If λ Multiplied to vector AB, then the magnitude is multiplied |λ| by and direction remain same (or opp.) according as λ is the + ue or, ue-

Unit Vectors

For a given vector a, the vector $\hat{a}=\frac{a}{|a|}$ gives the unit vector in the direction of a. for eg , if a=5i, then $\hat{a}=\frac{5i}{|5|}=i$,which is a unit vector.

Position Vectors

Position vector of a point P (x, y, z) is $x\hat{i}+y\hat{j}+z\hat{k}$ and its magnitude is $OP(r)=\sqrt{x^2+y^2+z^2}$ .For eg: Position vector of P (2,3,5) is $2\hat{i}+3\hat{j}+5\hat{k}$ and its magnitude is $\sqrt{2^2+3^2+5^2}=\sqrt{38}$.

Position of vectors

The Position vector of a point R dividing a line segment joining P,Q whose position vectors are a, b resp., in the ratio $m:n$
(i) internally is $\frac{na+mb}{m+n}$, (ii) externally is $\frac{mb-na}{m-n}$

Scalar Product of two vectors

If a, b are the vectors and 'θ', angle between them, then their scalar product $a.b=|a||b|\cos\theta$
$\Rightarrow \cos\theta=\frac{a.b}{|a||b|}$

Direction ration and direction cosines

The scalar components of a vector are its direction ratios, and represent its projections along the respective axes.

The magnitude (r) direction ratios (a,b,c) and direction cosines (l,m,n) of vector $a\hat{i}+b\hat{j}+c\hat{k}$ are related as:

$l=\frac{a}{r}, m=\frac{b}{r}, n=\frac{c}{r}$

For eg : If $\overline{AB}=\hat{i}+2\hat{j}+3\hat{k}$, then $r=\sqrt{1+4+9}=\sqrt{14}$

Direction ratios are (1,2,3) and direction cosines are $\left(\sqrt{\frac{1}{14}},\sqrt{\frac{2}{14}},\sqrt{\frac{3}{14}}\right)$

Cross Product of two vectors

$a\times b=|a||b|\sin\theta\,\hat{n}$, $\hat{n}$ is a unit vector perpendicular to line joining a,b.

Types of Vectors

(i) Zero vector (initial and terminal points coincide)
(ii) Unit vector (magnitude is unity.
(iii) Coinitial vectors (same initial points)
(iv) Collinear vectors (parellel to the same line)
(v) Equal vectors (same magnitude and direction)
(vi) Negative of a vector (same magnitude, opp. direction)

Properties of Vectors

If we have two vectors $a=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$, $b=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ and λ is any scalar, then-

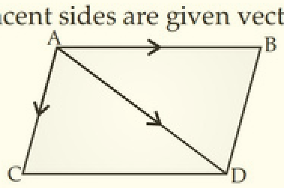
$a+b=(a_1+b_1)\hat{i}+(a_2+b_2)\hat{j}+(a_3+b_3)\hat{k}$

$\lambda a=(\lambda a_1)\hat{i}+(\lambda a_2)\hat{j}+(\lambda a_3)\hat{k}$

$a.b=a_1b_1+a_2b_2+a_3b_3$ and

$$a\times b=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\end{vmatrix}$$

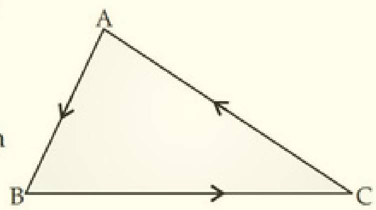
The vector sum of two coinitial vectors is given by the diagonal of the parallelogram whose adjacent sides are given vectors.



if $\overline{AB}, \overline{AC}$ are the given vectors, then $\overline{AB}+\overline{AC}=\overline{AD}$

The vector sum of the three sides of a triangle taken in order is 0 .i.e

if ABC is given triangle, then $\overline{AB}+\overline{BC}+\overline{CA}=0$.



Class : 12th Maths
Chapter- 11 : Three dimensional Geometry

Three dimensional Geometry

Shortest distance between

(i) two skew lines is the line segment perpendicular to both the lines

(ii) $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\mu\vec{b}_2$ is $\left|\dfrac{(\vec{b}_1\times\vec{b}_2)\cdot(\vec{a}_2-\vec{a}_1)}{|\vec{b}_1\times\vec{b}_2|}\right|$

(iii) $\dfrac{x-x_1}{a_1}=\dfrac{y-y_1}{b_1}=\dfrac{z-z_1}{c_1}$ and $\dfrac{x-x_2}{a_2}=\dfrac{y-y_2}{b_2}=\dfrac{z-z_2}{c_2}$ is $\dfrac{\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1c_2-b_2c_1)^2+(c_1a_2-c_2a_1)^2+(a_1b_2-a_2b_1)^2}}$

(iv) Parallel line $\vec{r}=\vec{a}_1+\lambda\vec{b}$ and $\vec{r}=\vec{a}_2+\mu\vec{b}$ is $\dfrac{\vec{b}\times(\vec{a}_2-\vec{a}_1)}{|\vec{b}|}$

Direction ratios and direction cosines of line

D. Cs of a line are the cosines of the angles made by the line with the positive direction of the co-ordinate axes. If l, m, n are the D. Cs of a line, then $l^2+m^2+n^2=1$. D. Cs of a line joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are $\dfrac{x_2-x_1}{PQ}, \dfrac{y_2-y_1}{PQ}, \dfrac{z_2-z_1}{PQ}$, where $PQ=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$

D.Rs of a line are the no.s which are proportional to the D.Cs of the line if l, m, n are the D.Cs and a, b, c are D.Rs of a line, then

$l=\dfrac{a}{\sqrt{a^2+b^2+c^2}}$, $m=\dfrac{b}{\sqrt{a^2+b^2+c^2}}$, $n=\dfrac{c}{\sqrt{a^2+b^2+c^2}}$

Equation of a plane

(i) which is at distance 'd' from origin and D.C.s of the normal to the plane as l,m,n is $lx+my+nz=d$.

(ii) $\perp r$ to a given line with D.Rs. A,B,C and passing through (x_1,y_1,z_1) is $A(x-x_1)+B(y-y_1)+C(z-z_1)=0$

(iii) Passing through three non-collinear points (x_1,y_1,z_1) (x_2,y_2,z_2), (x_3,y_3,z_3) is $\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix}=0$

Vector equation of a plane

(i) which contains three non-collinear points having position vectors $\vec{a}, \vec{b}, \vec{c}$ is $(\vec{r}-\vec{a})\cdot[(\vec{b}-\vec{a})\times(\vec{c}-\vec{a})]=0$.

(ii) That passes through the intersection of planes $\vec{r}\cdot\vec{n}_1=d_1$ & $\vec{r}\cdot\vec{n}_2=d_2$ is $\vec{r}\cdot(\vec{n}_1+\lambda\vec{n}_2)=d_1+\lambda d_2$, λ – non-zero constant.

Skew lines

These are the lines in space which are neither parallel nor intersecting. They lie in different planes. Angle between skew lines is the angle between two intersecting lines drawn from any point (origin) parallel to each of the skew lines.

Angle between the two lines

if l_1, m_1, n_1, l_2, m_2, n_2 are the D.Cs and $a_1, b_1, c_1, a_2, b_2, c_2$ are the D.Rs of the two lines and 'θ' is the acute angle between them, then $\cos\theta=|l_1l_2+m_1m_2+n_1n_2|=\left|\dfrac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}\right|$

Characteristics of planes

Two lines $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$, $\vec{r}=\vec{a}_2+\mu\vec{b}_2$ are coplanar if $(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)=0$. Equation of a plane that cuts co-ordinate axes at (a,0,0), (0,b,0), (0,0,c) is $\dfrac{x}{a}+\dfrac{y}{b}+\dfrac{z}{c}=1$.

The distance of a point with postion vector $\vec{a}$ from the plane $\vec{r}\cdot\hat{n}=d$ is $|d-\vec{a}\cdot\hat{n}|$. The distance from a point (x_1, y_1, z_1) to the plane $Ax+By+Cz+D=0$ is $\left|\dfrac{Ax_1+By_1+Cz_1+D}{\sqrt{A^2+B^2+C^2}}\right|$

Line of vector equations in 3D

Vector equation of a line passing through the given point whose position vector is $\vec{a}$ and parallel to a given vector $\vec{b}$ is $\vec{r}=\vec{a}+\lambda\vec{b}$

Vector equation of a line which passes through two points whose position vectors are $\vec{a}$ and $\vec{b}$ is $\vec{r}=\vec{a}+\lambda(\vec{b}-\vec{a})$

Equation of a line through point (x_1, y_1, z_1) and having D.Cs l, m, n is $\dfrac{x-x_1}{l}=\dfrac{y-y_1}{m}=\dfrac{z-z_1}{n}$ Also, equation of a line that passes through two points.

Angle between two lines

If 'θ' is the acute angle between $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$, $\vec{r}=\vec{a}_2+\lambda\vec{b}_2$ then, $\cos\theta=\left|\dfrac{\vec{b}_1\cdot\vec{b}_2}{|\vec{b}_1||\vec{b}_2|}\right|$

if $\dfrac{x-x_1}{l_1}=\dfrac{y-y_1}{m_1}=\dfrac{z-z_1}{n_1}$ and $\dfrac{x-x_2}{l_2}=\dfrac{y-y_2}{m_2}=\dfrac{z-z_2}{n_2}$ are the equations of two lines, then acute angle between them is $\cos\theta=|l_1\cdot l_2+m_1\cdot m_2+n_1\cdot n_2|$

Class : 12th Maths
Chapter- 12 : Linear Programming

Linear Programming

Fundamental Theorems

Theorem I : Let be the feasible region (convex polygon) for a R L.P. and let be the objective function. When has an Z = ax + by Z optimal value (max. or min.), where the variables are subject x ,y to the constraints described by linear inequalities, this optimal value must occur at a corner point (vertex) of the feasible region, Theorem 2: Let be the feasible region for a L.P.P, and let R Z= ax + by be the objective function. If is bounded then the O.F has R Z both a max. and a min. value on and each of these occurs at a R corner point (vertex) of . If the feasible region is unbounded, R then a max. or a min. may not exist. If it exists, it must occur at a corner point of R.

Definition

A. L.P.P is one that is concerned with finding the optimal value (max. or min.) of a linear function of several variables (called objective function) subject to the conditions that the variables are non- negative and satisty a set of linear inequalities (called linear constraints). Variables are sometimes called decision variables and are non- negative.

Types of L.P.P

(i) Diet problems
(ii) Manufacturing Problem
(iii) Transportation Problems

Corner point method

(i)Find the feasible region of the L.P.P and determine its corner points (vertices). (ii) Evaluate the O.F. at each corner Z = ax+by point. Let and be the largest and smallest values respectively M m at these points. If the feasible region is unbounded, and are M m the maximum and minimum values of the O.F. If the feasible region is unbounded , then (i) ' ' is the max. value of the O.F., if the open M half plane determined by has no point in common with ax+by>M the feasible region. Otherwise, the O.F. has no maximum value. (ii) ' ' is the minimum value of the O.F., if the open half plane m determined by has no point in common with the feasible ax+by

Solution of L.P.P

The common region determined by all the constraints including the non- negative constraint of a L.P.P is called the feasible region (or solution region) for the problem. Points within and on the boundary of the feasible region represent feasible solutions of the constraints. Any point outside the feasible region is an infeasible solution. Any point in the feasible region that gives the optimal value (max. or min.) of the objective function is called an optimal solution.

For Eg : Max Z = 250x + 75y, subject to the
Constraints: $5x + y \le 100$
$x + y \le 60$
$x \ge y\ 0,\ y \ge 0$ is an L.P.P

Probability

Random Variable

Real valued function whose domain is the sample space of a random experiment

The probability distribution of a random variable x is the system of numbers x: $x_1 \quad x_2 \quad . . . . , x_n$

$P(x)$: $p_1 \quad p_2 \quad . . . . \; p_n$ where , $pi>0$,

$\sum_{i=1}^{n} p_i = 1, \quad i=1.2,...., \quad n.$

Conditional probability

The probability of the event E is called the conditional probability of E given that F has already occurred, and is denoted by $P(E/F)$. Also,

$$P(E/F)=\frac{P(E\cap F)}{P(F)}, P(F)\neq 0.$$

Properties

(i) $0 \le P(E/F) \le 1,\ P(E'/F)=1-P(E/F)$

(ii) $P((E\cup F)/G)=P(E/G)+P(F/G)-P((E\cap F)/G)$

(iii) $P(E\cap F)=P(E)P(F/E), P(E)\neq 0$

(iv) $P(E\cap F)=P(F)P(E/F), P(F)\neq 0$

For *eg* : if $P(A)=\frac{7}{13},\ P(B)=\frac{9}{13}$ and $P(A\cap B)=\frac{4}{13}$, then

$$P(A/B)=\frac{p(A\cap B)}{P(B)}=\frac{\frac{4}{13}}{\frac{9}{13}}=\frac{4}{9}.$$

If E and F are independent, then P E $\cap$ F = P (E)P (F), P (E|F) = P (E), P (F) $\neq$ 0 and (F|F) = P(F), P(E) $\neq$ 0.

Variance and standard deviation

Let x be a R.V. whose possible values $x_1, x_2, ..., x_n$ occurs with probabilities $p(x_1), p(x_2), ... , p(x_n)$ respectively. Let, $\mu = E(x)$ be the mean of x. The variance of x, var (x) or $\sigma_x^2 = \sum_{i=1}^{n}(x_i-\mu)^2 p(x_i)$ or $E(x-\mu)^2$

The non-negative number

$6x=\sqrt{\mathrm{var}(x)}=\sqrt{\sum_{i=1}^{n}(x_i-\mu)^2 p(x_i)}$ is called the standard deviation of the R.V. 'X'. Also,

$\mathrm{var}(x)=E(x^2)-[E(x)]^2$ For *eg*: $E(x) = 3$ and $E(x^2) = 10$, then var $x = 10-9 = 1$ and SD$=\sqrt{1} = 1$.

Bernoulli's Trials

Trials of a random experiment are called Bernoulli trials, if they satisfy the following conditions :

(i)There should be a finite no. of trials.

(ii)The trials should be independent.

(iii)Each trial has exactly two outcomes: success or failure.

(iv)The probability of success remains the same in each trial.

For Binomial distribution, B (n,p), $P(X=x)=\left({}^nC_x q^{n-x} p^x, x=0,1,....,n\right)$ $(q=1-p)$

Theorem of total Probability

Let, $\{E_1, E_2 \; ....E_n\}$ be a partition of a sample space 'S' and suppose that each of $E_1, E_2, ...E_n$ has non-zero probability. Let 'A' be any event associated with S, then $P(A) = P(E_1)P(A|E_1) + P(E_2) P(A|E_2) + ... + P(E_n) P(A|E_n)$.

Bayes' Theorem

If $E_1, E_2, ..., E_n$ are events which constitute a partition of sample space S, *i.e.*, $E_1, E_2, . . . E_n$ are pair wise dis joint and $E_1\cup E_2\cup ....\cup E_n=S$ and A be any event with non-zero probability,

then $$P(E_i|A)=\frac{P(E_i)P(A|E_i)}{\sum_{j=1}^{n}P(E_j)P(A|E_j)}$$